XRPL Validator — Daily Engineering Report · 2026-03-29

Period: 2026-03-29 00:00 to 24:00 US Eastern (2026-03-29 04:00 to 2026-03-30 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: git history only. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 5 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat: 47 consecutive state hash matches — validator independently verifies XRPL** (`b7d73cf`, xrpl-validator) Major fixes that achieved consistent state hash matching against the network: - Sort transactions by TransactionIndex (execution order != array order) Root cause: OfferCancel at exec#19 appeared at array#46, OfferCreate at exec#21 at array#15 — processing in wrong order caused silent state drift - Add FeeSettings singleton to force-fetch list (4 protocol singletons total) - Handle entryNotFound for modified objects …
- **fix: LedgerHashes sub-page key computed dynamically (seq/65536)** (`cd280e4`, xrpl-validator) The XRPL state tree has a LedgerHashes sub-page that changes every ledger (new hash appended). Key = SHA512Half(0x0073 || uint32_be(seq/65536)). This object was never in AffectedNodes and wasn't being fetched, causing a persistent 32-byte drift after ~30-200 ledgers.
- **perf: branch-cached hash computation + skip second download** (`71c6f35`, xrpl-validator)
- Cache 16 root-level branch hashes, only recompute dirty branches - Fast path now marks dirty branches (was the cause of optimized-mode drift) - Skip second download entirely — ws_sync starts from pinned ledger - Skip-ahead when >50 ledgers behind (prevents rippled pruning stalls) - Dynamic LedgerHashes sub-page key: SHA512Half(0x0073 || seq/65536)
- **perf: single-pass merge + spawn_blocking — every ledger verified, feed stays smooth** (`5cf2901`, xrpl-validator)
- Replace O(k*n) Vec inserts with O(n) single-pass merge: update drops from 6-9s to 0.4s - Total hash time 2.8s — under the 3.5s ledger interval, now verifying every ledger - Move update_and_hash to spawn_blocking: SSE feed, peers, and web UI no longer freeze - Faster metadata retry bail-out: 3 attempts then break batch (prevents stale-ledger stall) - Partial failure tolerance: apply successful fetches even if some …

Docs and operations

- **license: replace MIT with proprietary — permission required for use** (`5ec2dfe`, xrpl-validator)

Independence program

No independence record for UTC day 2026-03-29; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
03-29 05:32	—	xrpl-validator	`b7d73cf`	feat: 47 consecutive state hash matches — validator independently verifies XRPL
03-29 14:49	—	xrpl-validator	`cd280e4`	fix: LedgerHashes sub-page key computed dynamically (seq/65536)
03-29 14:59	—	xrpl-validator	`5ec2dfe`	license: replace MIT with proprietary — permission required for use
03-29 19:34	—	xrpl-validator	`71c6f35`	perf: branch-cached hash computation + skip second download
03-30 03:19	—	xrpl-validator	`5cf2901`	perf: single-pass merge + spawn_blocking — every ledger verified, feed stays smooth

Times in UTC. A commit appears on the day it was committed to main or merged into it.